REALISING TWO INVERSE SYSTEMS OF DERIVED OBJECTS BY A BI-FILTERED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the statement is of the form 'In the situation of Lemma ...', so the statement of that lemma (source lines 2601-2614) is delivered as a context span with role=context to keep the card self-contained.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Realisation of one inverse system by a filtered complex. Same chapter, Lemma lemma-represent-by-filtered-complex, delivered on this card as a context span. Supplies the situation of the statement and the existence of a filtered complex realising the first inverse system, which the proof applies twice.
- Products in the derived category. Same chapter, Lemma lemma-derived-products. Supplies that the product of the two complexes realises the object, which is what allows the two filtrations to be combined on a single complex.

Source reference. Original definition 012L: sources/stacks-homology/pinned-original-source.tex, source lines 6175–6175. Navigation pointer invoked at proof line 2611; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BKI: sources/stacks-injectives/pinned-original-source.tex, source lines 2325–2325. Navigation pointer invoked at proof line 2719; the counted span contains the pointer only and the cited result is not reproduced

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Source reference. Original theorem 079P: sources/stacks-injectives/pinned-original-source.tex, source lines 2024–2024. Navigation pointer invoked at proof line 2552; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07D9: sources/stacks-injectives/pinned-original-source.tex, source lines 2226–2226. Navigation pointer invoked at proof line 2699; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Suppose given an object $E \in D(\mathcal{A})$ and an inverse system $\{E^i\}_{i\in\mathbf{Z}}$ of objects of $D(\mathcal{A})$ over $\mathbf{Z}$ together with a compatible system of maps $E^i \to E$. Picture:*

$$\ldots \to E^{i+1} \to E^i \to E^{i-1} \to \ldots \to E$$

Then there exists a filtered complex $K^\bullet$ of $\mathcal{A}$ (Homology, Definition 012L) such that $K^\bullet$ represents E and $F^iK^\bullet$ represents E^i compatibly with the given maps.

Lemma 0.2. *In the situation of Lemma 0.1 assume we have a second inverse system $\{(E')^i\}_{i\in\mathbf{Z}}$ and a compatible system of maps $(E')^i \to E$. Then there exists a bi-filtered complex $K^\bullet$ of $\mathcal{A}$ such that $K^\bullet$ represents E, $F^iK^\bullet$ represents E^i, and $(F')^iK^\bullet$ represents $(E')^i$ compatibly with the given maps.*

Proof. Using the lemma we can first choose $K^\bullet$ and F. Then we can choose $(K')^\bullet$ and F' which work for $\{(E')^i\}_{i\in\mathbf{Z}}$ and the maps $(E')^i \to E$. Using Lemma 0BKI we can assume $K^\bullet$ is a K-injective complex. Then we can choose a map of complexes $(K')^\bullet \to K^\bullet$ corresponding to the given identifications $(K')^\bullet \cong E \cong K^\bullet$. We can additionally choose a termwise injective map $(K')^\bullet \to J^\bullet$ with $J^\bullet$ acyclic and K-injective. (To do this first map $(K')^\bullet$ to the cone on the identity and then apply Theorem 079P.) Then $(K')^\bullet \to K^\bullet \times J^\bullet$ and $K^\bullet \to K^\bullet \times J^\bullet$ are both termwise injective and quasi-isomorphisms (as the product represents E by Lemma 07D9). Then we can simply take the images of the filtrations on $K^\bullet$ and $(K')^\bullet$ under these maps to conclude. □